

Harmonic Measure and Sets of Dimension

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Harmonic functions are averages

A continuous function $u : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is **harmonic** if its value at each point X is the average of its values on every sphere centered at X :

$$u(X) = \int_{\partial B_X(r)} u(Y) d\sigma(Y) \quad \text{for every ball } \overline{B_X(r)} \subset \Omega$$

Theorem

u is harmonic if and only if u is C^2 and $\Delta u \equiv u_{x_1x_1} + u_{x_2x_2} + \cdots + u_{x_nx_n} = 0$ throughout Ω

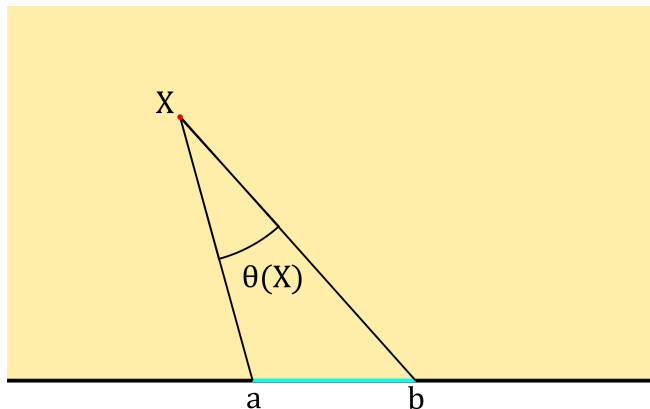
Examples

- ▶ ($n = 1$): harmonic functions on an interval are linear, $u(x) = cx + d$
- ▶ ($n = 2$): $u(x, y) = \sin(x)e^y$ and $u(x, y) = x^3 - 3xy^2$ are harmonic on $\mathbb{R}^2$

An example in the upper half plane

Let $\Omega = \{X = (x, y) : y > 0\}$ be the upper half plane

Let (a, b) be an interval on x -axis



The angle function $\theta(X) = \arctan \left(\frac{(b-a)y}{(x-a)(x-b)+y^2} \right) = \operatorname{Im} \log \left(\frac{x+iy-a}{x+iy-b} \right)$ is harmonic in Ω .

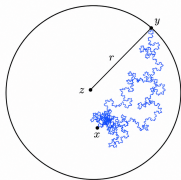
Averaging over other shapes

What if X is not the center of the ball, or the region is not a ball at all?

Theorem

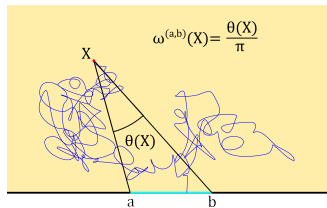
Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and let $X \in \Omega$. There is a probability measure ω^X on $\partial\Omega$, the **harmonic measure** of Ω with pole X , such that

$$u(X) = \int_{\partial\Omega} u(Y) d\omega^X(Y) \quad \text{for every } u \text{ harmonic in } \Omega \text{ and continuous on } \overline{\Omega}$$



$$\text{Ball } B_Z(r): d\omega^X = \frac{r^{n-2}(r^2 - |X - Z|^2)}{|X - Y|^n} \frac{d\sigma(Y)}{\sigma(\partial B_Z(r))}$$

(uniform when $X = Z$)



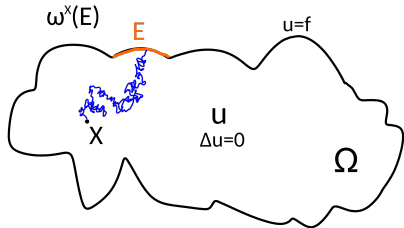
$$\text{Upper half plane: } \omega^X((a, b)) = \frac{\theta(X)}{\pi}$$

Harmonic measure is where Brownian motion exits

Start a random path (**Brownian motion**) at $X \in \Omega$. It first leaves Ω at a random point $B_{\text{exit}} \in \partial\Omega$, and

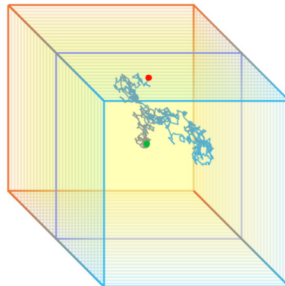
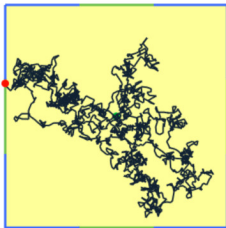
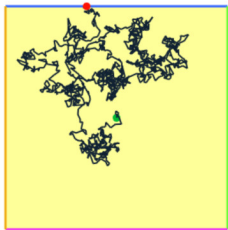
$$\omega^X(E) = \mathbf{Prob}(B_{\text{exit}} \in E : B_0 = X)$$

“Probability that a random path started at X first hits $\partial\Omega$ in E ”



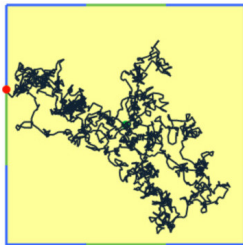
- ▶ Started at the center of a ball, Brownian motion exits uniformly by symmetry. This is the mean value property!
- ▶ **Dirichlet problem:** for boundary data f , the average $u_f(X) = \int_{\partial\Omega} f(Y) d\omega^X(Y)$ is harmonic in Ω and has boundary values f **except on a set of harmonic measure zero**
- ▶ This is one reason to be interested in the null sets and the dimension of harmonic measure

Brownian motion / harmonic measure demos



Live demos: brownian-motion-demos.html

Brownian motion / harmonic measure demos



Answer: The **green edges** E are the slight favorite. Using separation of variables, one can find a Fourier sine series solution for the harmonic measure and evaluate it numerically:

$$\omega^{\text{center}}(E) = 4 \sum_{n=1}^{\infty} \frac{2}{n\pi} \frac{\sinh(n\pi/2)}{\sinh n\pi} (\cos(n\pi/3) - \cos(2n\pi/3)) \sin(n\pi/2) \approx 0.522873\dots$$

Physical interpretations

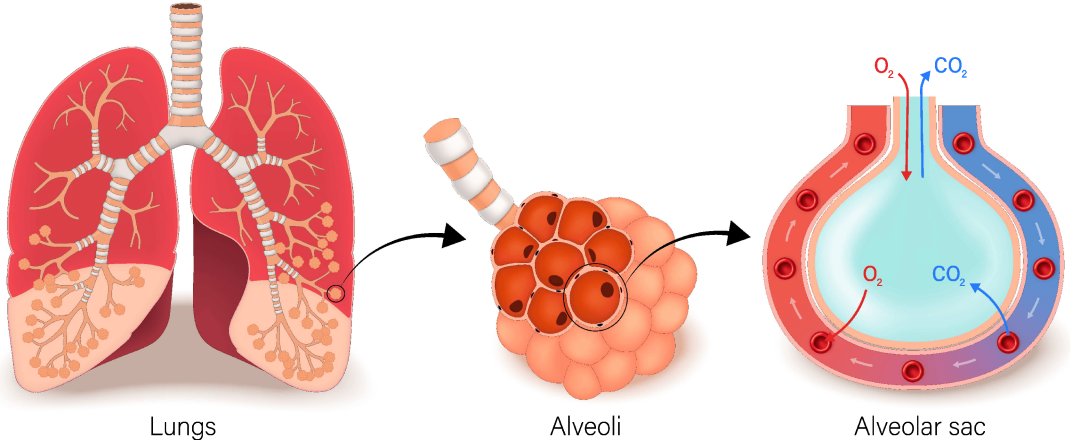
Diffusion describes the net movement of molecules from regions of higher concentration to regions of lower concentration.

Brownian motion models the trajectory of a single molecule undergoing diffusion

Examples of physical phenomena controlled by diffusion

- ▶ Heat transfer
- ▶ Electric charge on a metallic surface
- ▶ Passivation / fouling / poisoning / restricted absorption
- ▶ Transfer of nutrients from digestive system to blood
- ▶ Transfer of oxygen from air to blood during respiration

HUMAN RESPIRATORY SYSTEM



$\approx 480,000,000$ alveoli in human lungs

Alveolus / Alveolar Sac has Diameter $\approx 200\text{-}500\ \mu m$

Blood-Air Interface has Diameter $\approx 0.5\ \mu m$

Oxygen Atom has Diameter $\approx 0.0003\ \mu m$

Passivation of irregular surfaces (Filoche et al.)

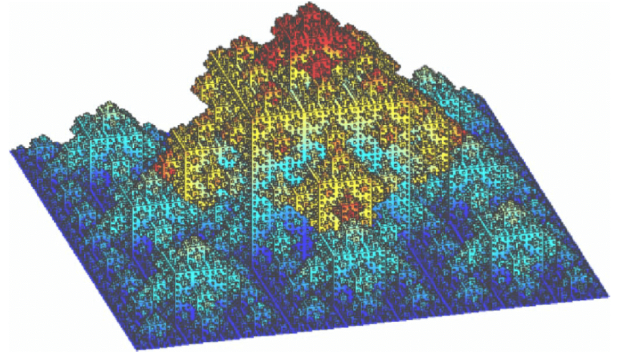
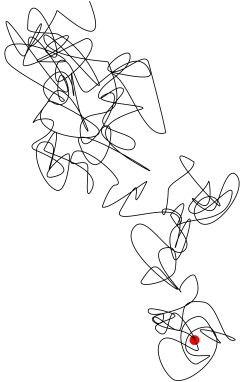
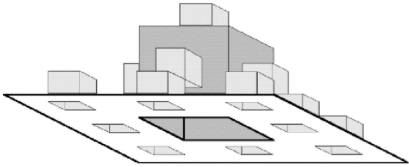


Fig. 3. Successive active regions during the passivation process of a fifth-generation prefractal surface based on the cubic Koch surface. Diffusing particles are coming from below and reach first mainly the blue region of the interface. Each color in the simulation represents a set of four successive passivated regions (dark blue corresponds to regions 1–4, light blue to 5–8, etc.). One can see that the size of the active region gradually decreases during the passivation process. At the end, only the dark-red regions on the tip are active.

Interlude: What is dimension?



3d Reconstruction of Human Lungs from CT-scans

Andreas Heinemann (CC BY 2.5), Wikimedia

Cantor sets

For $0 < \lambda < 1/2$, build $C(\lambda)$ by repeatedly keeping the two outer intervals of relative length λ from each interval:

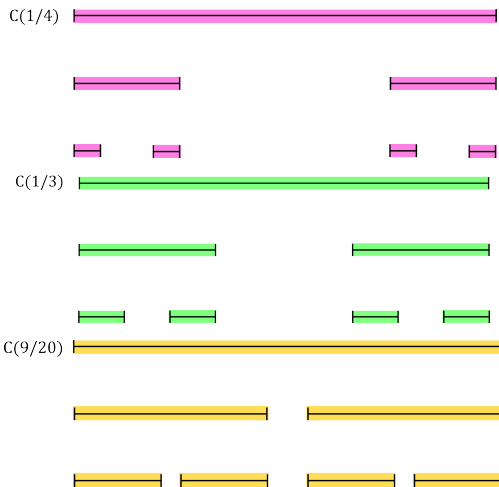
$$C(\lambda) = \bigcap_{k=0}^{\infty} C_k(\lambda)$$

- ▶ $C_k(\lambda)$ has 2^k intervals of length λ^k
- ▶ Total length of $C_k(\lambda)$ is $(2\lambda)^k \rightarrow 0$

Length cannot distinguish $C(1/4)$,
 $C(1/3)$, $C(9/20)$

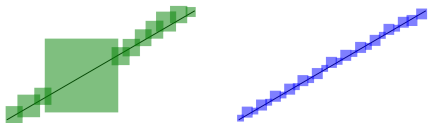
But our intuition says that

$C(1/4) < C(1/3) < C(9/20)$



Hausdorff content and Hausdorff dimension

Cover $A \subseteq \mathbb{R}^n$ by cubes of any size. A cube of side length ℓ costs ℓ^s . Find the cheapest cover:

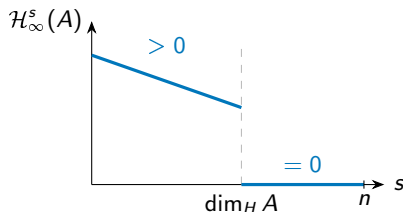


$$\mathcal{H}_{\infty}^s(A) := \inf \left\{ \sum_i (\text{side } Q_i)^s : A \subset \bigcup_i Q_i \right\}$$

$\mathcal{H}_{\infty}^s$ is the (cubical) **s -dimensional Hausdorff content**

There is a unique number $d \in [0, n]$ such that $\mathcal{H}_{\infty}^s(A) > 0$ for all $s < d$ and $\mathcal{H}_{\infty}^s(A) = 0$ for all $s > d$. This number is the **Hausdorff dimension** $\dim_H A$.

Same null sets and same dimension as the Hausdorff measure $\mathcal{H}^s = \lim_{\delta \rightarrow 0} \mathcal{H}_{\delta}^s$, which only allows cubes of side length at most δ



Hausdorff dimension of Cantor sets

For all $\lambda \in (0, 1/2)$, the Cantor set $C(\lambda)$ has Hausdorff dimension

$$\dim_H C(\lambda) = \frac{\log(2)}{\log(1/\lambda)} \in (0, 1)$$

Why ($\leq$): covering $C(\lambda)$ by the 2^k intervals of $C_k(\lambda)$ costs $2^k \lambda^{ks} = (2\lambda^s)^k$,
which tends to 0 as $k \rightarrow \infty$ whenever $2\lambda^s < 1$, i.e., whenever $s > \log(2)/\log(1/\lambda)$



$$\dim_H C(1/4) = 0.500 \dots < \dim_H C(1/3) = 0.630 \dots < \dim_H C(9/20) = 0.868 \dots$$

Dimension of human lungs

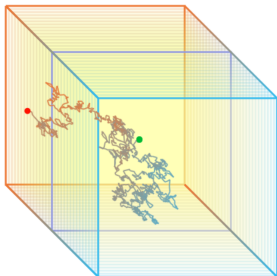
Estimates of “Fractal Dimension” in the Literature



Approximation of scaling laws for different measurements of lungs along branching pathways (airway length, airway diameter) from experimental data leads to:

- ▶ Nelson and Manchester (1988): dimension is ≈ 2.64 and ≈ 2.76
- ▶ Nelson, West, and Golden (1990): dimension is ≈ 2.26 and ≈ 2.4
- ▶ Weibel (1991): dimension is ≈ 2.35
- ▶ Lamrini Uahabi and Atounti (2017): dimension is ≈ 2.88 .

Landing sets and the dimension of harmonic measure



Let $\Omega \subset \mathbb{R}^n$ be bounded, open, connected. A set $L \subset \partial\Omega$ is a **landing set for Brownian motion** if a.e. random path started in Ω first hits the boundary in L : $\omega^X(\partial\Omega \setminus L) = 0$

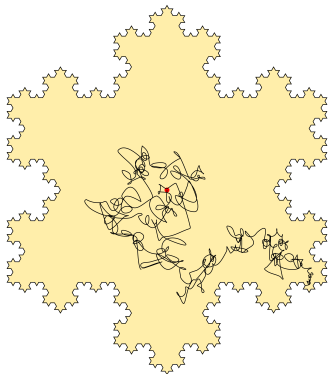
E.g. The 6 *open faces* of the cube $(0, 1)^3$ (exclude the edges) form a landing set

The **Hausdorff dimension of harmonic measure on Ω** is the smallest Hausdorff dimension of a landing set for Brownian motion in Ω

The **Hausdorff dimension of harmonic measure on $\mathbb{R}^n$** is the largest Hausdorff dimension of harmonic measure on Ω across all domains:

$$d_n = \sup_{\Omega} \overline{\dim}_H \omega_{\Omega}, \quad d_n \geq n - 1 \text{ (e.g. take } \Omega = (0, 1)^n \text{)}$$

Dimension drop on the snowflake



$$\dim_H \partial\Omega = \log_3(4) = 1.2618\dots$$

Three theorems from 1985

- ▶ **Kaufman and Wu:** on the snowflake domain, there is a landing set L with $\dim_H L < \dim_H \partial\Omega$. We call this phenomenon **dimension drop**
- ▶ **Carleson:** on the snowflake domain, there is a landing set L with $\dim_H L = 1$
- ▶ **Makarov:** on every simply connected planar domain, there is a landing set L with $\dim_H L = 1$, and every set $E \subset \partial\Omega$ with $\dim_H E < 1$ has $\omega_\Omega(E) = 0$

Brownian motion does not see the whole fractal

What do we know about d_n ?

Recall $n - 1 \leq d_n \leq n$.

- ▶ **Jones and Wolff 1988:** every planar domain has a landing set of dimension at most 1. Thus, $d_2 = 1$.
- ▶ **Wolff 1995:** there is a domain $\Omega \subset \mathbb{R}^3$ with $\overline{\dim}_H \omega_\Omega > 2$. Thus, $d_n > n - 1$ for all $n \geq 3$.
- ▶ **Bourgain 1987:** $d_n < n$ for all $n \geq 3$.

Neither Wolff nor Bourgain says by how much

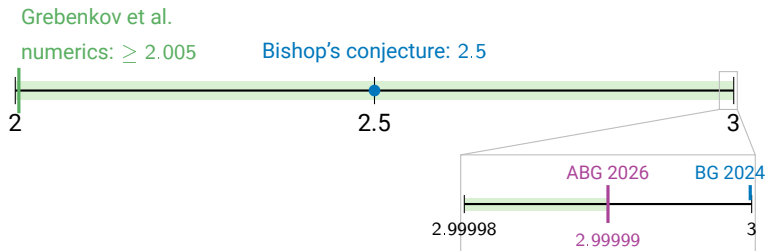
Conjecture (Bishop 1992)

The dimension of harmonic measure in $\mathbb{R}^n$ is $d_n = n - \frac{1}{n-1}$.

In particular, $d_3 = 2.5$.

Today: new upper bounds on d_n and challenges that could be stepping stones towards Bishop's conjecture. **We need more people to get involved!**

What is the dimension of harmonic measure in $\mathbb{R}^3$?



Theorem (Badger–Genschaw 2024)

$$d_3 < 2.99999\ 99999\ 99999$$

$$d_n < n - \left(\frac{1}{n-1}\right)^{2n^2} \text{ provided that } n \gg 3$$

Theorem (Albert–Badger–Genschaw 2026)

$$d_3 < 2.99999$$

$$d_n < n - \left(\frac{1}{n-1}\right)^{2n+6.12\sqrt{n}} \text{ for all } n \geq 3$$

On the scale $n - d_n \approx \left(\frac{1}{n-1}\right)^E$, the exponent improves from $E = 2n^2$ to $E = 2n + O(\sqrt{n})$.

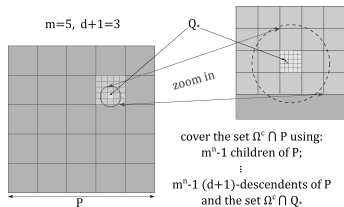
Bishop's conjecture is $E = 1$.

Bourgain's strategy: a dichotomy at every scale

Fix $s = n - \rho$ slightly less than n . Look at the boundary $\partial\Omega$ inside of each cube P .

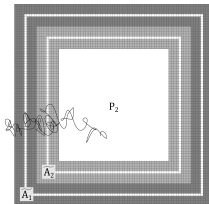
Alternative I: there is a hole

Then $\partial\Omega \cap P$ can be covered by smaller cubes Q_i efficiently: $\sum_i (\text{side } Q_i)^s \leq (\text{side } P)^s$



Alternative II: there is no hole

Then $\partial\Omega$ acts as a barrier. Brownian motion is unlikely to reach deep inside P , which is quantified by a constant λ



If one of the alternatives holds for every cube, then $\overline{\dim}_H \omega_\Omega \leq n - \frac{\lambda\rho}{1+\lambda}$

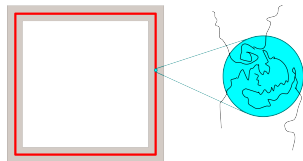
(Pictures: the model of Badger–Genschaw with a fixed m -adic grid)

What is new: one ring is enough

- ▶ Previous versions of Alternative II used **many concentric rings** of cubes in a fixed m -adic grid, with m large depending on n
- ▶ **One-ring alternative:** if one thin ring around a cube has no large hole, then Brownian motion is unlikely to enter the inside of the cube before exiting the domain
- ▶ Brownian motion can **return to the ring many times**, and every return is another chance to exit the domain

Other ingredients:

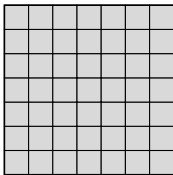
- ▶ pairwise disjoint cubes in **general position, mixed-scale covers** (next slide), and **round holes**
- ▶ **approximate polar integration:** a nearly sharp bound on the potential $\int |Y - X|^{-(n-2)} d\mu(Y)$ when $\mu(Q) \leq (\text{side } Q)^s$ for all cubes Q



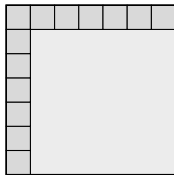
The gains are cumulative: five decimals in $\mathbb{R}^3$ needs all of them together

A hole helps only if it is big enough

When $s < n$, bigger cubes are more efficient: splitting a cube into smaller cubes **increases** $\sum_i (\text{side } Q_i)^s$, even though the total volume stays the same. So a hole must be big enough to pay for the split.



Uniform cover: a hole helps only if
 $n - s \lesssim (\text{volume of the hole}) / \log m$



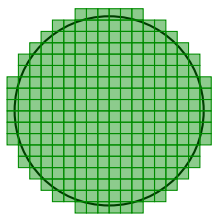
Mixed cover: a hole helps as soon as
 $n - s \leq (\text{volume of the hole}) / (n + 1)$

Principle

A hole in a cube leads to an efficient cover when $n - s$ is (a bit) smaller than the relative volume of the hole. Covers using cubes of mixed sizes make "a bit" much less costly.

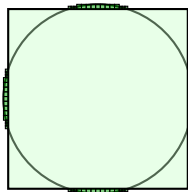
An elementary open problem

What is the s -dimensional (cubical) Hausdorff content $\mathcal{H}_\infty^s(B(0, 1))$ of the unit ball?



256 squares of side 0.12

$$\mathcal{H}_\infty^{1.75}(B) \leq 6.2633\dots$$



1 square of side 1.9, plus 60
squares of sides 0.05 and 0.03

$$\mathcal{H}_\infty^{1.75}(B) \leq 3.3170\dots$$

One square of side 2 gives

$$\mathcal{H}_\infty^{1.75}(B) \leq 2^{1.75} = 3.3635\dots$$

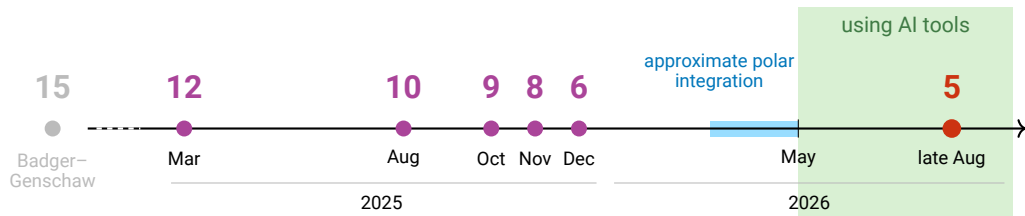
Optimizing uniform grids gives

$$\mathcal{H}_\infty^s(B) \leq \omega_n \frac{n^n}{s^s} \left(\frac{\sqrt{n}}{n-s} \right)^{n-s}$$

Open Problem

Compute $\mathcal{H}_\infty^s(B(0, 1))$ exactly for any $n - 1 < s < n$. (To our knowledge, the exact value is unknown even in the plane.)

How we got to five decimals



- ▶ Each number counts the decimals in the proved bound $d_3 < 2.99 \dots 9$
- ▶ The principal ideas come from more than two years of research and predate our use of AI tools
- ▶ The steps from 15 down to 6 decimals came before we used AI tools
- ▶ The last step, from 6 to 5 decimals together with the exponent $2n$ for all n , came after we started using AI tools

How we used AI

Tools: Claude Opus 5, Claude Fable 5.1, and GPT 5.6 Sol

What they did:

- ▶ symbolic and numerical computation
- ▶ scripts for searches over the discrete parameters of the method, e.g. the ring scale M and the collar scale L
- ▶ high-precision and interval-arithmetic checks of the numerical certificates
- ▶ quickly testing variants of the method, most of which we rejected (e.g. asymmetric rings, multiple holes)
- ▶ figures and manuscript preparation

What did not change:

- ▶ **Everything is proved unconditionally** and verified by the authors. Each numerical bound comes from explicit parameters substituted into proved inequalities, and can be checked with a calculator.

We could have reached these results eventually, but AI tools compressed months of computation and writing into weeks

Ten challenges on the dimension of harmonic measure

In the paper, we propose ten intermediate challenges to spur further research:

1. **Numerical Experiment:** estimate $\overline{\dim}_H \omega_\Omega$ numerically for explicit domains in $\mathbb{R}^3$
2. **Explicit Upper Bound:** prove $b_3 \leq w < 1$ for an explicit w
3. **One Decimal:** prove $b_3 \geq 0.1$ or $b_3 \leq 0.9$
4. **Exact Dimension:** find $\Omega \subset \mathbb{R}^3$ with $\overline{\dim}_H \omega_\Omega > 2$ and compute it exactly
5. **Growing Excess:** dimension excess that grows with n
6. **Exponential Bound:** prove $b_n \geq 10^{-Cn}$
7. **Packing Dimension:** is there a packing dimension drop?
8. **Maximal Dimension Drop:** how large can $\dim_H \partial\Omega - \overline{\dim}_H \omega_\Omega$ be?
9. **Denjoy Transition:** when does dimension drop begin for Cantor sets in a line?
10. **Elliptic Bourgain:** Bourgain's theorem for elliptic measures

► remaining challenges

One Decimal Challenge

Bishop's Conjecture: The Hausdorff dimension of harmonic measure in $\mathbb{R}^3$ is 2.5

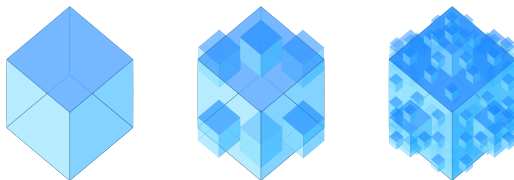
Current Bounds: $2.005 \overset{?}{\leq} d_3 < 2.99999$

Challenge 3 (One Decimal)

For trivial reasons, $d_3 \leq 2.9$ or $d_3 \geq 2.1$, but we don't know the truth of either bound. Prove one of these (or both)!

- ▶ In our view, a resolution of Challenge 3 would be an accomplishment on par with determining the value of d_3
- ▶ Even an explicit lower bound is open: find a domain in $\mathbb{R}^3$ with dimension of harmonic measure $> 2 + \varepsilon$ for some explicit $\varepsilon > 0$ (Challenge 2)

Numerical Experiment Challenge



- ▶ Grebenkov et al. (2005) simulated Brownian motion on the fifth stage of the cubical Koch snowflake G^3 and estimated that the dimension of harmonic measure is ≥ 2.005 (not yet mathematically verified)
- ▶ In $\mathbb{R}^n$, each face of G^n is replaced by $3^{n-1} + 2n - 2$ faces of relative size one-third, so $\overline{\dim}_H \omega_{G^n} \leq \log_3(3^{n-1} + 2n - 2) \rightarrow n - 1$. In our view, this is indirect evidence that $\overline{\dim}_H \omega_{G^3}$ is much smaller than d_3 .

Challenge 1 (Numerical Experiment)

Repeat the experiment to confirm or refute the 2.005 bound. Estimate the dimension of harmonic measure on other domains in $\mathbb{R}^3$, e.g. domains whose boundaries are cubical Cantor sets.

Exponential Bound Challenge

Bishop's Conjecture: The Hausdorff dimension of harmonic measure in $\mathbb{R}^n$ is $n - \frac{1}{n-1}$

Current Upper Bound: $d_n < n - \left(\frac{1}{n-1}\right)^{2n+6.12\sqrt{n}}$ for all $n \geq 3$

Proposition (Albert, Badger, and Genschaw 2026)

No choice of parameters in our model of Bourgain's alternative gives a better lower bound on $n - d_n$ than $\left(\frac{1}{n-1}\right)^{2n}$, up to lower order terms in the exponent.

Challenge 6 (Exponential Bound)

Prove that $d_n \leq n - 10^{-Cn}$ for all $n \geq n_0$ for some constants C and n_0 or prove any sub-exponential lower bound on $n - d_n$ when n is large.

- Challenge 6 likely needs a new idea beyond Bourgain's alternative

ROBIN GREEN FUNCTION ESTIMATES AND A MODEL OF MAMMALIAN LUNGS

GUY DAVID, STEFANO DECIO, MAX ENGELSTEIN, MARCEL FILOCHE, SVITLANA MAYBORODA AND MARCO MICHETTI

ABSTRACT. The present paper establishes delicate properties of the Green function with Robin boundary conditions, in particular, elucidating the nature of the passage between the Dirichlet-like and Neumann-like behaviour. This yields sharp quantifiable bounds on the corresponding harmonic measure and proves the phase transition in the behavior of the total flow earlier conjectured in physics literature in concert with the efficacy of mammalian lungs.

1. INTRODUCTION

We continue our study of the Robin problem in rough domains, started in [DDEMM24]. Recall the Robin problem, with parameter $a \in (0, \infty)$ in a domain $\Omega \subset \mathbb{R}^n$

$$(1.1) \quad \begin{aligned} -\operatorname{div}(A(x)\nabla u) &= 0, & x \in \Omega \\ \frac{1}{a}A(Q) \cdot \nabla u(Q) \cdot \nu(Q) + u(Q) &= f(Q) & Q \in \partial\Omega, \end{aligned}$$

(where $\nu(Q)$ is the outward pointing unit normal to $\partial\Omega$ at Q). As a goes from 0 to ∞ the boundary condition (suitably renormalized) seems to interpolate between the well-studied Dirichlet and Neumann boundary conditions. Yet, the properties of the Dirichlet and Neumann solutions near the boundary are so different that understanding and quantifying this passage is a challenging task. Partially motivated by applications to the study of mammalian lungs, we embarked on this journey in [DDEMM24]. Conjectures in the biomedical engineering community predicted a phase transition in the efficacy of a lung and hence, likely, in the dimension of the harmonic measure with the Robin boundary conditions depending on the value of the parameter a . Expectations based on the mathematics of the classical Dirichlet harmonic measure also seemed to indicate that the Robin problem should exhibit

Today I talked about classical harmonic measure, but there are many interesting relatives which are being actively investigated:

elliptic measures, p -harmonic measures, caloric measure, parabolic measures, harmonic measures in higher co-dimensions, Robin harmonic measures, ...

Thank you for your attention

Appendix: dimension lemma, cubes in general position

Theorem (Theorem 4.1)

Let μ be a Borel regular outer measure on $\mathbb{R}^n$ that is finite on bounded sets, and let $E \subset \mathbb{R}^n$ be a Borel set with $\mu(\mathbb{R}^n \setminus E) = 0$. Suppose that there exist parameters $\alpha \in (0, 1)$, $\rho \in (0, n)$, $\beta \in (0, 1)$, $\tau \in (1, \infty)$, and $\gamma \in (0, 1)$ such that for every cube P in $\mathbb{R}^n$ with side $P \leq 1$, at least one of the following conditions holds:

- (I) there is a cover of $E \cap P$ by pairwise disjoint cubes $Q_1, Q_2, \dots$ contained in P with side $Q_i \leq \beta$ side P such that

$$\sum_i (\text{side } Q_i)^{n-\rho} \leq (\text{side } P)^{n-\rho}$$

- (II) there is a cover of $E \cap P$ by pairwise disjoint cubes $R_1, R_2, \dots$ contained in P with side $R_i \leq \beta$ side P such that

$$\sum_i (\text{side } R_i)^{n-\rho} \leq \tau (\text{side } P)^{n-\rho} \quad \text{and} \quad \sum_i \mu(R_i)^\alpha (\text{vol } R_i)^{1-\alpha} \leq \gamma \mu(P)^\alpha (\text{vol } P)^{1-\alpha}$$

Then the upper Hausdorff dimension of μ is at most

$$n - \left(\frac{\lambda_\alpha}{1 + \lambda_\alpha} \right) \rho, \quad \text{where } \lambda_\alpha := \frac{-\log_\tau(\gamma)}{1 - \alpha} \in (0, \infty).$$

Appendix: the one-ring alternative (Theorem 6.1)

◀ What is new

Let ω_n denote the volume of the unit ball in $\mathbb{R}^n$. Let $\rho_0 := 0.2/(n^6 \log h)$ and $\eta := \frac{2a}{n+n-5}(1 - 2^{-(n-2)})$.

Theorem

Let $n \geq 3$ and let $0 < a < 1$. Let $M \geq 3$, $L \geq 2$, and $h \geq n^3$ be integers with $2L \geq 3M$ such that

$$W := \left(1 + \frac{n\omega_n}{\omega_{n-1}} \left(\frac{4L}{3M} - 1\right) \eta\right)^{-1} < \left(1 - \frac{2}{M}\right)^n =: V,$$

and let $\rho \in (0, \rho_0]$ satisfy

$$\begin{aligned} & \left(1 - \frac{1}{L}\right)^{n-\rho} + (L^n - (L-1)^n - 2^{n-1}) \left(\frac{1}{L}\right)^{n-\rho} + 2^{n-1}(2^n - 1) \left(\frac{1}{2L}\right)^{n-\rho} \\ & + \left(2^{n-1}h^n - \frac{\omega_n}{2^n} (h - 2\sqrt{n})^n\right) \left(\frac{1}{2Lh}\right)^{n-\rho} + a\omega_n \left(\frac{1}{4L}\right)^{n-\rho} \leq 1. \end{aligned}$$

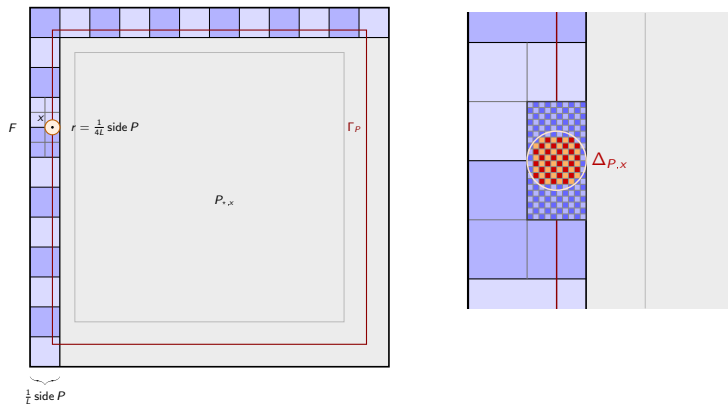
Then $d_n \leq n - \lambda\rho/(1 + \lambda)$, where

$$\lambda := W \log_\tau \frac{W}{V} + (1 - W) \log_\tau \frac{1 - W}{1 - V}, \quad \tau := \left(1 - \frac{2}{M}\right)^{n-\rho} + (M^n - (M-2)^n) \left(\frac{1}{M}\right)^{n-\rho}.$$

The five terms count the big cube, the medium cubes, the little cubes, the tiny cubes near the hole, and the hole.

Appendix: the partition used in Alternative I

◀ Bourgain's strategy



Illustrated with $n = 2$, $M = 8$, $L = 12$, and $h = 10$. The 'L'-shaped collar is divided into medium cubes. Only those near the ball $B_{P,x}$ are subdivided into little cubes, and only the little cubes covering $B_{P,x}$ are refined into tiny cubes.

Appendix: parameters for the lower bounds on b_n

[◀ Main results](#)

Certificates (Theorem 7.2, proved)

n	M	L	a	h	$b_n \geq$
3	9	17	117/200	10^5	1.0118×10^{-5}
4	10	17	13/20	10^5	1.206×10^{-7}
5	11	19	143/200	10^6	8.006×10^{-10}

Plug these into Theorem 6.1 and check the ρ -condition with a calculator, at $\rho = 1.05628 \times 10^{-5}$, 1.22747×10^{-7} , 8.0084×10^{-10}

Tuned values (computed, not certified)

n	M	L	a	h	$b_n \geq$	c_n
6	13	22	0.750	10^6	3.530×10^{-12}	2.731
7	14	25	0.795	10^6	1.166×10^{-14}	2.558
8	16	29	0.800	10^6	2.910×10^{-17}	2.446
9	18	32	0.855	10^6	5.856×10^{-20}	2.366
10	18	37	0.840	10^6	9.128×10^{-23}	2.310
50	109	264	0.970	50^6	3.521×10^{-166}	1.958
100	254	654	0.990	100^6	4.429×10^{-384}	1.921

Computed at 60–80 significant digits. Each value equals $(n-1)^{-c_n n}$, and c_n decreases toward 2.

For comparison, Bishop's conjecture is $b_n = (n-1)^{-1}$

Appendix: remaining challenges (1/2)

◀ Ten challenges

Challenge 4 (Exact Dimension)

Construct a domain $\Omega \subset \mathbb{R}^3$ such that $\overline{\dim}_H \omega_\Omega > 2$ and compute $\overline{\dim}_H \omega_\Omega$ exactly.

Wolff constant: $w_\Omega := \overline{\dim}_H \omega_\Omega - (n - 1)$ and $w_n := \sup_{\Omega \subset \mathbb{R}^n} w_\Omega = 1 - b_n$.

Challenge 5 (Growing Excess)

Construct a family of domains $\Omega^n \subset \mathbb{R}^n$, for $n \geq 3$, arising from a single geometric construction, such that $0 < w_{\Omega^3} < w_{\Omega^4} < w_{\Omega^5} < \dots$. (Trivial increases from products $\Omega \times \mathbb{R}$ do not count.)

Packing analogue of Bourgain's constant: $p_n := n - \sup_{\Omega \subset \mathbb{R}^n} \overline{\dim}_P \omega_\Omega$. Since packing dimension dominates Hausdorff dimension, $0 \leq p_n \leq b_n$.

Challenge 7 (Packing Dimension)

Prove or disprove that $p_n > 0$ for every $n \geq 2$. If $p_n > 0$, is $p_n = b_n$? In particular, is $p_2 = 1$?

Appendix: remaining challenges (2/2)

◀ Ten challenges

Maximal dimension drop constant: $m_n := \sup_{\Omega \subset \mathbb{R}^n} (\dim_H \partial\Omega - \overline{\dim}_H \omega_\Omega) \geq b_n$.

Challenge 8 (Maximal Dimension Drop)

For each $n \geq 2$, is $m_n = b_n$ or is $m_n > b_n$? If $m_2 > 1$, compute m_2 or obtain nontrivial bounds for it.

Challenge 9 (Denjoy Transition)

Determine the least $s_0 \in (0, 1)$ such that, for every $s \in (s_0, 1)$ and every s -Ahlfors regular compact set $E \subset \mathbb{R}$, the harmonic measure of $\mathbb{R}^2 \setminus E$ satisfies $\overline{\dim}_H \omega_{\mathbb{R}^2 \setminus E} < s$. Is there an analogous transition for s -Ahlfors regular compact sets $E \subset \mathbb{R}^2$ not confined to a line?

Elliptic Bourgain constant: $b_n(\Lambda) := n - \sup_{\Omega, A} \overline{\dim}_H \omega_\Omega^A$ over domains Ω and Λ -elliptic symmetric coefficients A , so $b_n(1) = b_n$.

Challenge 10 (Elliptic Bourgain)

Prove that $b_n(\Lambda) > 0$ for every $n \geq 3$ and every $1 \leq \Lambda < \infty$, and determine the rate at which $b_n(\Lambda) \rightarrow 0$ as $\Lambda \rightarrow \infty$ for each $n \geq 2$.